

1. Algorithm

1. Start

2. Read two positive integers `m` and `n`.

3. Check if `m > n` or `m ≤ 0` or `n ≤ 0`. If true, display error message and stop.

4. Declare integer variable `choice` and set it to `0`.

5. Start a `while` loop that runs while `choice != 5`:

- Display menu options (1 to 5).
- Read user choice into `choice`.
- Use `switch(choice)`:
 - **Case 1:** Set `i = m`. While `i ≤ n`, display multiples of `i` from 1 to 10 using inner `while` loop with counter `j`. Increment `i`.
 - **Case 2:** Set `i = m`. While `i ≤ n`, display addition table for `i (i + j)` from `j = 1` to 10. Increment `i`.
 - **Case 3:** Set `i = m`. While `i ≤ n`, display subtraction table for `i (i - j)` from `j = 1` to 10. Increment `i`.
 - **Case 4:** Set `i = m`. While `i ≤ n`, display division results `((float)i / j)` formatted to 2 decimal places from `j = 1` to 10. Increment `i`.
 - **Case 5:** Print exit message.
 - **Default:** Print "Invalid choice! Please enter a number between 1 and 5."

6. Stop

2. Test-Case Table

S.No.	Input (m, n)	Menu Choice	Expected Output	Remarks
1	m = 5, n = 3	N/A	Invalid Input! Make sure $m \leq n$ and both are positive.	Validates input condition $m \leq n$
2	m = 2, n = 3	1	Prints 10 multiples for 2 (2 to 20) and 3 (3 to 30)	Multiples Operation
3	m = 2, n = 3	2	Addition table for 2 (2+1 to 2+10) and 3 (3+1 to 3+10)	Addition Operation
4	m = 2, n = 3	3	Subtraction table for 2 (2-1 to 2-10) and 3 (3-1 to 3-10)	Subtraction Operation
5	m = 2, n = 3	4	Division table for 2 and 3 by 1..10 (e.g. $2/1=2.00$, $2/3=0.67$)	Division Operation (2 decimals)
6	m = 2, n = 3	8	Invalid choice! Please enter a number between 1 and 5.	Invalid Option Handling
7	m = 2, n = 3	5	Exiting program. Goodbye!	Program Exit

3. C Source Code

```
#include <stdio.h>

int main() {
    int m, n;
    int choice = 0;
    int i, j;

    printf("Enter starting positive integer (m): ");
    scanf("%d", &m);
    printf("Enter ending positive integer (n): ");
    scanf("%d", &n);

    if (m <= 0 || n <= 0 || m > n) {
        printf("Invalid Input! Make sure m <= n and both are positive integers.
");
        return 0;
    }

    while (choice != 5) {
        printf("
===== MENU =====
");
        printf("1. Print first 10 multiples
");
        printf("2. Print addition table (1 to 10)
");
    }
}
```

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");
    printf("3. Print subtraction table (1 to 10)
");
    printf("4. Print division table (1 to 10)
");
    printf("5. Exit
");
    printf("Enter your choice (1-5): ");
    scanf("%d", &choice);

    switch (choice) {
        case 1:
            i = m;
            while (i <= n) {
                printf("
--- Multiples of %d ---
", i);

                j = 1;
                while (j <= 10) {
                    printf("%d x %d = %d
", i, j, i * j);

                    j = j + 1;
                }
                i = i + 1;
            }
            break;

        case 2:
            i = m;
            while (i <= n) {
                printf("
--- Addition Table for %d ---
", i);

                j = 1;
                while (j <= 10) {
                    printf("%d + %d = %d
", i, j, i + j);

                    j = j + 1;
                }
                i = i + 1;
            }
            break;

        case 3:
            i = m;
            while (i <= n) {
                printf("
--- Subtraction Table for %d ---
", i);

                j = 1;
                while (j <= 10) {
                    printf("%d - %d = %d
", i, j, i - j);

                    j = j + 1;
                }
            }
            break;
    }
}

```

```

        }
        i = i + 1;
    }
    break;

    case 4:
        i = m;
        while (i <= n) {
            printf("
--- Division Table for %d ---
", i);

            j = 1;
            while (j <= 10) {
                printf("%d / %d = %.2f
", i, j, (float)i / j);
                j = j + 1;
            }
            i = i + 1;
        }
        break;

    case 5:
        printf("
Exiting program. Goodbye!
");
        break;

    default:
        printf("
Invalid choice! Please enter a number between 1 and 5.
");
        break;
    }
}

return 0;
}

```